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Boundary-Spanning Activity and Employee Reactions: An Empirical Study[1]

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Boundary-spanning activity was studied in a large manufacturing company through a sample of 192 managers, engineers, and supervisors. Contrary to prior theory and research, this study found boundary-spanning activity unrelated to role conflict or ambiguity and positively related to job satisfaction for the total sample. Boundary-spanning activity was also positively related to a number of job characteristics for the total sample. Marked differences in boundary-spanning activity and its relationships with other variables, however, were found across occupational levels. While managers and engineers generally had boundary-spanning activity related to high levels of job satisfaction and job characteristics, first-level supervisors had boundary-spanning activity related to higher role conflict and lower job satisfaction with opportunities for promotion.

Organizational theory and research in the last decade have been strongly directed toward an open-system approach. The modern organization is now being viewed as a system with permeable boundaries, trying to adapt to an uncertain and changing environment. A number of writers (Miller & Rice, 1967; Thompson, 1967; Aldrich, 1971; Litterer, 1973) have focused on the importance of boundary processes which enable organizations to monitor, sense, and interact with environmental forces. The purpose of the present re-

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search was to study the relationships between boundary-spanning activity (BSA) and role and satisfaction variables in a large manufacturing company. In addition, this study also investigated the relationships between BSA and job characteristics in order to gain some insight into the underlying job factors which may be affected by BSA.

PRIOR RESEARCH

BSA, which may be defined as the interpersonal transfer of information across organizational boundaries, has been found to be vital for effective organizational functioning by research into a number of diverse organizations. Pruden (1969) and Pruden and Reese (1972) studied outside salesmen in a wood products company; Wren (1967) examined an aerospace project and an electric power pool; and Allen and Cohen (1969), Allen, Piepmeier, and Cooney (1971), and Organ and Greene (1972) studied research and development organizations. All these studies concluded that BSA plays an important role in maintaining organizational viability.

Thompson's (1967) conceptual framework also places considerable importance on boundary-spanning components and BSA for organizational viability. BSA is seen to occur most often at the higher levels of the organization where management is interacting with the environment, while those employees at lower levels in the technical core are usually buffered from environmental factors. Thompson also sees the nature of the environment as an important influence on the type of BSA that occurs. When the environment is homogeneous and stable, BSA can be standardized and often can become a routine activity. With a heterogeneous and shifting environment, however, those individuals with high BSA have more important positions and more discretion in decision making.

Much of the prior empirical research, however, has found negative effects from BSA. Kahn, Wolfe, Quinn, Snoek, and Rosenthal (1964) developed a theory of role dynamics which proposes that BSA leads to role pressures and tensions. Since both internal and external role senders have expectations for performance for those who engage in BSA, inconsistent expectations may result in role conflict, and vague or unclear expectations may result in role ambiguity. A survey of 725 people representing the national labor force in the United States, and an intensive study of 53 subjects selected from seven industrial organizations showed that those engaged in a high degree of BSA experienced substantial role conflict and tension, and often had role senders who misunderstood the requirements of engaging in BSA. The Kahn et al. data therefore support their theory that having role senders in more than one organization results in strong and conflicting role pressures.

Organ and Greene (1972) also tested the Kahn et al. theory in two samples of research and development managers who differed on characteristics associated with boundary roles. They found the managers with boundary-role characteristics to have lower role consensus (agreement on role expectations between the position incumbent and superior), and lower role compliance (agreement on actual role performance and superior's expectations) than those without boundary-role characteristics. In addition, lower correlations between sentiments and performance evaluation, as well as between role compliance and job satisfaction, were found for the managers with boundary-role characteristics. The Organ and Greene findings therefore added further support to the Kahn et al. theory.

Laboratory experiments have also focused on the negative effects of BSA. Organ (1971) conducted a bargaining experiment and found that higher visibility of BSA leads to greater experienced tension and conformity to norms. Wall and Adams (1974) experimented with negotiations by salesmen and buyers and found that obedience to constituents by those engaging in BSA resulted in more trust and positive evaluations. These experiments again point out the important and often negative effects of BSA.

Much of the research that investigates job characteristics has focused on the relationship between job characteristics and employee attitudes and behavior. Particular emphasis has been placed on job design and on the moderating effects of individual differences in the job characteristic/employee attitudes relationship (see Hackman & Lawler, 1971; Wanous, 1974; Sims, Szilagyi, & Keller, 1976). Thompson (1967) has pointed out that the job provides the individual with an arena or sphere of action in which to meet the demands placed on him by the social system. Thompson defines spheres of action in terms of the dimensions of the opportunity to learn (e.g., variety, autonomy, feedback, task identity) and the opportunity for visibility (e.g., interaction, dealing with others). With respect to the aspects of boundary spanning, Thompson postulated that boundary-spanning jobs would not only offer greater spheres of action, but that these jobs would vary considerably depending on the degree to which the environment at the boundary is homogeneous or heterogeneous, stable or shifting.

The Kahn et al. (1964) theory of role dynamics and Thompson's (1967) conceptual framework were used in the present study to generate hypotheses regarding BSA. In the Kahn et al. theory, organizational, interpersonal, and personality factors were posited as affecting role expectations, pressures, and tensions. Perceptions of inconsistent and vague expectations by the role incumbent tend to result in role conflict and ambiguity, and, consequently, coping responses to conflict and ambiguity are posited to include low job satisfaction, role rejection, or withdrawal. Thompson sees BSA affecting the nature of job characteristics for those who engage in this activity.

Basically, then, the Kahn et al. (1964) theory focuses on role tensions from BSA, while the Thompson (1967) framework sees BSA leading to a broader and more varied range of job activities. While potentially inconsistent, these two approaches enable one to research BSA with a wide range of important variables.

> **Hypothesis 1:** BSA will be positively related to role conflict and ambiguity.
> **Hypothesis 2:** BSA will be negatively related to job satisfaction.
> **Hypothesis 3:** BSA will be positively related to the job characteristics of variety, autonomy, task identity, feedback, dealings with others, and friendship opportunities.

The rationale for the first hypothesis was that conflicting and vague role expectations from role senders both internal and external to the company's boundaries will result in higher role conflict and ambiguity for those who engage in BSA. The second hypothesis was based on the response to role pressures in the form of lower job satisfaction. The rationale for the third hypothesis was that BSA provides the individual with greater spheres of action and discretion, and therefore should be positively related to the job characteristics of variety, autonomy, task identity, feedback, friendship opportunities, and dealings with others.

METHOD

Sample

The study was conducted in a large company located in the southwestern United States which manufactured capital equipment via large and small batch technologies. Sales for 1974 exceeded $200 million. The subjects were 192 managerial, engineering, and supervisory personnel (64% response rate) located at the company headquarters and main manufacturing plant. The average age of the subjects was 45.4 years, the average total work experience was 23.4 years, and the average time in present position was 6.6 years; 85% of the subjects had attended college.

The total sample was also broken down by occupational levels as follows: top and middle-level managers (N = 93), engineering personnel (N

= 48), and first-level supervisors (N = 33). Other respondents were 18 administrative personnel in various positions who could not be categorized in this manner and were not included in the analyses by occupational levels.

Questionnaires were distributed to the subjects by company mail, and returned directly to the researchers in postpaid envelopes by United States mail. Respondents were guaranteed confidentiality of information.

Measures

Boundary-Spanning Activity. Prior work by Creighton, Jolly, and Denning (1972) and Allen and Cohen (1969) had validated and successfully utilized items for the identification of boundary spanners in the information and technology transfer process in research and development organizations. These items were selected on the basis of concurrent validity with sociograms, and not the face validity or content of the particular items. A 4-item BSA scale based on this prior work was developed and validated in two earlier studies of boundary-spanning in a research and development organization (Keller & Holland, 1975a; 1975b). This BSA scale was used in the present study, with the BSA score being the sum of the 4 scale items.

To validate the scale for the present sample, management of the manufacturing company was asked to identify some of their employees who management thought were high on BSA (those who regularly exchanged information with personnel in outside organizations). All employees identified by company management as being high on BSA scored in the top third of the BSA scale distribution; moreover, a t-test between the means of the BSA scores of the identified boundary spanners and of the other employees was highly significant (t = 16.00, $p < .001$). Thus, the BSA scale showed concurrent validity in being able to identify employees high on BSA, which is consistent with prior research (Creighton et al., 1972; Allen & Cohen, 1969; Keller & Holland, 1975a; 1975b). The reliability (Coefficient Alpha) of the BSA scale was .81, with an average score of 10.54 and a standard deviation of 3.90. The items for the BSA scale and a correlation matrix are provided in the Appendix.

Role Conflict and Ambiguity. The scale developed by Rizzo, House, and Lirtzman (1970) was used to measure an individual's role perceptions. Role conflict (8 items) was defined as the perception of conflicting demands by the role incumbent. Role ambiugity (6 items) was defined as the lack of clarity or predictability one perceives in his work-related behavior. A factor analysis of the instrument for this sample confirmed the structure of each of the two role variables. In a separate analysis of the data presented in this paper, Szilagyi, Sims, and Keller (1976) showed that the role conflict and ambiguity scales had high factor congruencies across multiple samples, as computed by the Wrigley-Neuhaus procedure described in Harmon (1967). The split-half reliabilities (corrected by the Spearman-Brown formula) for this sample were 0.94 for role conflict and .90 for role ambiguity.

Job Satisfaction. The Job Description Index (JDI) was used to measure job satisfaction (Smith, Kendall & Hulin, 1969). According to Smith, Kendall, and Hulin, job satisfaction represents the difference between what is expected and what is experienced in relation to the alternatives available in a given situation. The JDI was intended to gauge the affective responses to this difference by measuring feelings associated with different facets of the job situation. Specifically, the JDI measures satisfaction with five aspects of a job: the work itself, supervision, coworkers, pay, and opportunities for promotion on the job. The validity and reliability of this instrument across different samples have been well established in the literature (Smith, Kendall & Hulin, 1969; Smith, Smith & Rollo, 1974).

Job Characteristics. Many of the questions in the job characteristics instrument were taken from the Hackman and Lawler (1971) research. In order to improve reliability, other questions which appeared to have face validity were added to each scale. A full description of the 35-item instrument, including extensive validity and reliability analyses across multiple samples, can be found in Sims, Szilagyi, and Keller (1976). Overall, the variables exceeded acceptable criteria for validity and reliability. The six job characteristic dimensions measured by the instrument were:

Variety: The degree to which a job requires employees to perform a wide range of operations in their work and/or the degree to which employees must use a variety of equipment and procedures in their work.

Autonomy: The extent to which employees have a major say in scheduling their work, selecting the equipment they will use, and deciding on procedures to be followed.

Task Identity: The extent to which employees do an entire or whole piece of work and can clearly identify the result of their efforts.

Feedback: The degree to which employees receive information as they are working which reveals how well they are performing on the job.

Dealing with Others: The degree to which a job requires employees to deal with other people to complete their work.

Friendship Opportunities: The degree to which a job allows employees to talk with one another on the job and to establish informal relationships with other employees at work.

Split-half reliabilities (corrected by the Spearman-Brown formula) for the sample in this study were as follows: 0.93 for variety, 0.93 for autonomy,

Table I. Zero-Order Correlations between Boundary-Spanning Activity and Role, Satisfaction, and Job Characteristic Variables (Total Sample, N=192) (Two-Tailed Tests of Significance)

	Role conflict	Role ambiguity	Dimensions of job satisfaction: Work	Pay	Supervision	Promotions	Coworkers
Boundary-spanning activity	.04	.08	.30[a]	.05	.07	.06	.14[b]

	Job characteristics: Variety	Autonomy	Task identity	Feedback	Friendship	Dealing with others
Boundary-spanning activity	.36[a]	.23[c]	.14[b]	.02	.25[a]	.14[b]

[a] $p < .001$.
[b] $p < .05$.
[c] $p < .01$.

0.90 for feedback, 0.88 for task identity, 0.89 for dealing with others, and 0.89 for friendship opportunities.

RESULTS

The correlational analysis performed on the total sample (N = 192) is reported in Table I. The analysis shows almost no relationships between BSA and role conflict and ambiguity; therefore, the first hypothesis was not supported. BSA was positively related to satisfaction with the work itself and with coworkers. These relationships were opposite to those of hypothesis 2; therefore, no support was given for the second hypothesis.

Striking relationships appeared between BSA and the job characteristics. Here, BSA was positively related to variety, autonomy, task identity, friendship opportunities, and dealing with others which were relationships predicted in hypothesis 3. To summarize, the first two research hypotheses were not supported by the data, while the third hypothesis was supported.

BSA was also analyzed across the three occupational groups by Duncan's multiple range test with Kramer's (1956) extension for unequal sample sizes, and the results are reported in Table II. It is clear that BSA increases directly with occupational level from supervisors to engineers to managers (Spearman rho = .35, $p < .001$). These results support Thompson's (1967) conceptual framework which sees upper level management

Table II. Duncan's Multiple Range Test with Kramer's Extension for Unequal Sample Sizes on Boundary-Spanning Activity by Occupational Level

Boundary-spanning activity	Occupational level: Managers (N = 93)	Engineers (N = 48)	Supervisors (N = 33)
Mean	11.28	9.79	7.69
Standard deviation	3.96	2.63	2.65
Differences between groups	← $p < .05$ →	← $p < .01$ →	
	← $p < .001$ →		

interacting most with the environment, while those in the technical core (first-level supervisors) are usually buffered from environmental factors.

Since BSA was significantly different for each occupational level, the data were also analyzed for each level using the three research hypotheses, and these results are presented in Table III. For the managerial group the first two research hypotheses were not supported, while the third hypothesis was supported: BSA was not related to role conflict or ambiguity; BSA was

Table III. Zero-Order Correlations between Boundary-Spanning Activity and Role, Satisfaction, and Job Characteristic Variables Broken Down by Occupational Level (Two-Tailed Tests of Significance)

Variable	Managers (N = 93)	Engineers (N = 48)	Supervisors (N = 33)
Role variables			
Role conflict	.13	−.04	.42[a]
Role ambiguity	.04	.18	.06
Job satisfactions			
Work	.24[a]	.32[a]	.08
Pay	.11	−.10	−.29
Supervision	.10	.19	−.22
Promotions	.17	.00	−.47[b]
Coworkers	.05	.28[a]	−.09
Job characteristics			
Variety	.45[c]	.28[a]	−.25
Autonomy	.38[b]	−.08	.09
Task identity	.10	−.28[a]	.32
Feedback	.02	.07	.00
Friendship	.27[b]	−.14	.28
Dealing with others	.25[a]	−.17	.22

[a] $p < .05$.
[b] $p < .01$.
[c] $p < .001$.

positively related to satisfaction with the work itself; and BSA was positively related to the job characteristics of variety, autonomy, friendship opportunities, and dealing with others. The same general relationships as those with the managers were found for the engineers with the exception that BSA was negatively related to task identity. The data for the supervisors, however, showed different relationships. Here, BSA was positively related to role conflict, and negatively related to satisfaction with opportunities for promotion. Some small support for the first two hypotheses can therefore be found from the results with the supervisors. For the managers and engineers, however, only the third hypothesis received support.

DISCUSSION

The results of this study are basically unsupportive of the Kahn et al. (1964) theoretical framework and much of the prior research relating to BSA. For the total sample, the data show BSA not related to high levels of role conflict and ambiguity. BSA was related, moreover, to higher satisfaction with the work itself and with coworkers, and to the job characteristics of variety, autonomy, task identity, friendship opportunities, and dealing with others. Rather than role pressures and low satisfaction, then, the total sample results suggest that positions with high levels of BSA are desirable and more satisfying for incumbents.

The data also indicated that BSA is a more important and desirable position component at higher occupational levels. Consistent with Thompson's (1967) conceptual framework, BSA increased with an increase in occupational level. At the lower level of supervision, moreover, BSA appears to take on some of the negative relationships hypothesized from the Kahn et al. (1964) theory. Here, higher role conflict and lower job satisfaction with opportunities for promotion are related to BSA. One should also note that the number of significant relationships between BSA and the other variables increased with occupation level; hence, BSA appears to be a more important variable for explaining behavioral relationships at higher levels of the organization. This pattern is also consistent with Thompson's framework. The data also suggest, as predicted by Thompson in hypothesis 3, that BSA itself may also increase the amount of variety, autonomy, task identity, friendship opportunities, and dealing with others in a job. Much of the prior theory and research relating to BSA has focused on its negative aspects without fully appreciating the potentially positive aspects that BSA can have for those who engage in this activity.

This study's finding that BSA is positively related to job satisfaction may possibly be explained with the concept of information as a power resource. Pettigrew (1972) found that those individuals with high levels of BSA were able to obtain a valuable organizational resource—power, in the form of information. Thompson (1967) and Galbraith (1973) also suggest that by permitting individuals to exercise discretion, boundary-spanning jobs enable individuals to reduce environmental uncertainties and contingencies, and, to the extent that these aspects are important to the organization, the individual gains power, particularly in a bargaining process. Since those who engage in BSA are able to obtain information from external sources not usually accessible to others, they become more powerful and important to the organization and can use this resource to obtain better and more satisfying jobs.

It may also be that we have been treating BSA as a global variable, and that different types of BSA actually occur in an organization with different consequences. The relative lack of power of lower-level supervisors may increase the conflict and dissatisfaction associated with BSA. Upper-level managers, however, may find that BSA gives them additional power vis-à-vis their peers, and hence it could be a desirable activity.

There is clearly a need for additional research into BSA. Research, perhaps longitudinal, is needed to examine the types of BSA that occur at different levels of an organization. Such research, moreover, should consider the interorganizational context within which the BSA occurs. This perspective is needed to help explain why BSA may have favorable or unfavorable effects for those who engage in the activity.

REFERENCES

ALDRICH, H. Organizational boundaries and inter-organizational conflict. *Human Relations,* 1971, *24,* 279-293.

ALLEN, T. J., & COHEN, S. I. Information flow in research and development laboratories. *Administrative Science Quarterly,* 1969, *14,* 12-19.

ALLEN, T. J., PIEPMEIER, J. M., & COONEY, S. The international technological gatekeeper. *Technology Review,* 1971, *73.*

CREIGHTON, J. W., JOLLY, J. A., & DENNING, S. A. *Enhancement of research and development output utilization efficiencies; linker control methodology in the technology transfer process.* Monterey, California: Naval Postgraduate School, June 30, 1972.

GALBRAITH, J. *Designing complex organizations.* Reading, Massachusetts: Addison-Wesley, 1973.

HACKMAN, J. R., & LAWLER, E. E. Employee reactions to job characteristics. *Journal of Applied Psychology,* 1971, *55,* 259-286.

HARMAN, H. H. *Modern factor analysis.* Chicago: University of Chicago Press, 1967.

KAHN, R. L., WOLFE, D. M., QUINN, R. P., SNOEK, J. D., & ROSENTHAL, R. A. *Organizational stress: Studies in role conflict and ambiguity.* New York: Wiley, 1964.

KELLER, R. T., & HOLLAND, W. E. Boundary-spanning roles in a research and development organization: An empirical investigation. *Academy of Management Journal,* 1975, *18,* 388-393. (a)

KELLER, R. T., & HOLLAND, W. E. Boundary-spanning activity and research and development management: A comparative study. *IEEE Transactions on Engineering Management,* 1975, *EM-22,* 130-133. (b)

KRAMER, C. Y. Extension of multiple range tests to group means with unequal numbers of replications. *Biometrics,* 1956, *12,* 307-310.

LITTERER, J. A. *The analysis of organizations.* Second edition. New York: Wiley, 1973.

MILLER, E. J., & RICE, A. K. *Systems of organization.* London: Tavistock, 1967.

ORGAN, D. W. Some variables affecting boundary role behavior. *Sociometry,* 1971, *34,* 524-537.

ORGAN, D. W., & GREENE, C. N. The boundary relevance of the project manager's job. *R & D Management,* 1972, *3,* 7-11.

PETTIGREW, A. M. Information control as a power resource. *Sociology,* 1972, *6,* 187-204.

PRUDEN, H. O. Interorganizational conflict, linkage, and exchange: A study of industrial salesmen. *Academy of Management Journal,* 1969, *12,* 339-350.

PRUDEN, H. O., & REESE, R. M. Interorganization role-set relations and the performance and satisfaction of industrial salesmen. *Administrative Science Quarterly,* 1972, *17,* 601-609.

RIZZO, J. R., HOUSE, R. J., & LIRTZMAN, S. I. Role conflict and ambiguity in complex organizations. *Administrative Science Quarterly,* 1970, *15,* 150-163.

SIMS, H. P., SZILAGYI, A. D., & KELLER, R. T. The measurement of job characteristics. *Academy of Management Journal,* 1976, in press.

SMITH, P. C., KENDALL, L. M., & HULIN, C. L. *The measurement of satisfaction in work and retirement.* Chicago: Rand McNally, 1969.

SMITH, P. C., SMITH, O. W., & ROLLO, J. Factor structure for blacks and whites of the job Description Index and its discrimination of job satisfaction. *Journal of Applied Psychology,* 1974, *59,* 99-100.

SZILAGYI, A. D., SIMS, H. P., & KELLER, R. T. Role dynamics, locus of control, and employee attitudes and behavior. *Academy of Management Journal,* 1976, in press.

THOMPSON, J. D. *Organizations in action.* New York: McGraw-Hill, 1967.

TURNER, A. N., & LAWRENCE, P. R. *Industrial jobs and the worker.* Boston: Harvard Graduate School of Business Administration, 1965.

WALL, J. A., & ADAMS, J. S. Some variables affecting a constituent's evaluations of and behavior toward a boundary role occupant. *Organizational Behavior and Human Performance,* 1974, *11,* 390-408.

WANOUS, J. P. Individual differences and reactions to job characteristics. *Journal of Applied Psychology,* 1974, *59,* 616-627.

WREN, D. A. Interface and interorganizational coordination. *Academy of Management Journal,* 1967, *10,* 69-81.

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APPENDIX

Boundary-Spanning Activity Scale

Each of the following items had a five-point scale of "0," "1–2," "3–4," "5–6," and "more than 6."

1. Indicate with a check the total number of magazines, journals, and newspapers which you regularly read.
2. During the last month, indicate with a check the frequency with which you recommended specific information sources (e.g., journal article, knowledgeable colleague, report, magazine, etc.) to a colleague.
3. During the last month, indicate with a check the frequency with which you have sought information/advice from members of other organizations outside (name of company).
4. During the last month, indicate with a check the frequency with which members of organizations outside (name of company) have sought information/advice *from* you.

Correlation Matrix of BSA Scale Items

Item	1	2	3	4
1		.60	.40	.40
2			.51	.51
3				.63